Hiten Dalmia

hitendwork@gmail.com | linkedin/hiten-dalmia | github/confumbit | confumbit.github.io (homepage)

Education

Chennai Mathematical Institute — Chennai, India
Master of Science in Theoretical Computer Science (On Full Scholarship) — *Aug. 2026 – Discontinued*

Kirori Mal College, University of Delhi — Delhi, India
Bachelor of Science (Honours with Research) in Mathematics, Minor in Economics — *Nov. 2022 – May 2026*

Experience

Data Scientist & Analytics Consultant — Mar. 2023 – Present
Freelance & Fractal — *Remote*

- Delivered 10+ web and analytics applications for clients via Upwork and Fractal, specializing in full-stack architecture, business intelligence reporting, and automated data pipelines.
- **Order Ingestion & Business Intelligence Solution** | *Python, SQL, PostgreSQL, Power BI, LLM*
  - Built a full-stack order portal using NextJS, ExpressJS, TypeScript, and PostgreSQL with session-based authentication and RESTful APIs to deliver business intelligence reporting.
  - Structured a dual-tier portal with customer ordering and an administrative suite with Power BI dashboards for real-time inventory tracking, demand forecasting, and executive KPIs.
  - Engineered an automated data pipeline in Python parsing WhatsApp messages into PostgreSQL, using an LLM framework fallback to process free-text orders across 1,000+ monthly transactions.
  - Formulated relational PostgreSQL schemas with indexed views and integrity constraints to execute high-volume SQL analytics transactions with sub-50ms query latency.

Computer Vision Engineer — Oct. 2023 – Apr. 2025
Tagglabs Experimental Pvt. Ltd. (Contract) — *Gurugram, Haryana (Remote)*

- **Cross-platform AR App** | *Python, Flutter, OpenCV, C++, Django* **(Live link)**
  - Architected a native C/C++ bridge using `dart:ffi` to port OpenCV SIFT algorithms into Flutter, enabling cross-platform computer vision and feature extraction across Android and iOS.
  - Constructed a real-time video camera feed data pipeline normalizing OS frame buffers, converting video streams into C structs with zero memory leaks at 30+ FPS performance.
  - Implemented real-time keypoint extraction and machine learning detection heuristics, transmitting match coordinates asynchronously to Dart for responsive augmented reality UI overlay rendering.
  - Developed mobile interfaces in Flutter from Figma designs and built RESTful backend APIs in Django to power location analytics and spatial discovery features for alpha release.
- **Kia National Conference App** | *Flutter, Power BI*
  - Designed an event analytics mobile app in Flutter for 500+ attendees at the Kia Dealer Conference, building real-time attendee tracking and Power BI dashboards for senior leadership.

Technical Skills

Data Science & Analytics: Python, SQL, R, Pandas, NumPy, Scikit-learn, Seaborn, Matplotlib, Power BI
AI & LLM Frameworks: LLM Frameworks (LangChain), LLM Context Engineering, Agentic Development
Data Engineering & Databases: PostgreSQL, Automated Data Pipelines, RESTful APIs, Node.js, ExpressJS
Developer Tools & Cloud: AWS, Git, Linux, Bash, Docker, NextJS, Flutter, React, LaTeX

Achievements

Erasmus Mundus Joint Master MATHS-DISC Fellowship — Europe
Issued by European Union and affiliated universities — *Mar. 2026*

Master in Mathematics (ISI) and MSc Computer Science (CMI) Scholarships — India
Indian Statistical Institute and Chennai Mathematical Institute respectively — *Jun. 2026*

Publications

Computable Analysis: Mathematical Foundations and Computability of Real Functions
Poincare Journal of Mathematical Physics & Applications, Vol. 1, No. 1 (2026), 1–20 (with Sujit Kumar)